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Energy storage integration with run of river power plants to mitigate operational environmental constraints: Case study of Sweden

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ABSTRACT

Increasing environmental restrictions, create stringent regulations for the use and management of rivers and water bodies, thereby jeopardizing the operational flexibility of hydropower plants. The national plan for new environmental regulations in Sweden is expected to be implemented in 2025. Hence, it is imperative for the hydropower producers to consider alternatives which can overcome the environmental restrictions for providing flexibility. In this paper, a Mixed Integer Linear Programming (MILP) based algorithm for the short-term regulation of hydropower plants with energy storage is proposed. There are two stages of the algorithm: Future electricity market prices model using machine learning techniques, and the combined hydropower plant and battery system (CHBS)'s operation model using MILP. The designed algorithm is used to analyze the techno-economic feasibility of the operation of three hydropower plants Skattungbyn, Unnån and Hansjö, located in lower Oreälven river in Sweden. Three different electricity market scenarios were developed based on Swedish nuclear energy targets for 2040. The forecasted market prices act as inputs for the designed optimization algorithm. The algorithm optimizes the operation of hydropower plants with an objective to maximize revenue through harnessing flexibility enabled by storage. It is observed from the results that with the current battery costs (≈0.36 million €/MWh), the CHBS for short-term regulation is not profitable with NPV ranging from −1 to −4 million € and the cost of battery needs to be less than 50 000 €/MWh to make it economically feasible.

1. Introduction

The industrial era's rapid increase in pollution and CO_2 content in the atmosphere has led to the increased greenhouse effect and global warming in recent decades. In addition to it, the depleting fossil fuel resources and increase in natural fuels prices like oil and gas demands to shift our focus on the renewable energy (RE) sources. It is also advantageous to diversify energy supply through different sources and reduce the dependencies on imported fuels. With the current growth rate in greenhouse gas emissions, climate change is expected to impact all habitats, including humans and all forms of flora and fauna. Following this, the Paris Agreement was signed by nations worldwide, and it mainly emphasizes decreasing global warming. One main aspect in that agreement was the targets of RE production set by every country to mitigate greenhouse gas emissions. The whole of the European Union has a target to achieve 32% RE production by the year 2030 [1]. Sweden reached its 2020 targets in the year 2012 and currently has a 54% RE share, whereas the European Union has a 17% RE share in the electricity mix [2]. The 2040 energy targets for Sweden are 100% RE share in the electricity mix [3].

With such ambitious targets, RE would be adapted extensively to be used to its maximum potential, but RE does come with some issues. RE being dependent on natural resources, is intermittent and has geographic limitations. One potential solution to the above posed drawbacks is integrating different sources of energy. Hybrid systems may increase the reliability of energy production than standalone systems. One of the most common forms of hybrid energy systems is integrating an energy storage system along with an energy-producing system. For example, the possibility of integrating a battery energy storage system (BESS) with hydropower plants enables it to have flexible electricity generation. The integration of such BESS would also avoid the environmental restrictions, and the combined system can be regulated based on the electricity market.

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In Sweden, hydropower is a major source of the RE in the energy system, in which the reservoir type power plant can regulate the production based on the fluctuations in the demand. It constitutes around 40% of the total electricity supplied over the years [4]. The licenses to operate hydropower plants in Sweden have some restrictions in terms of controlling the discharge and the license revision, which is expected to happen in the near future according to the EU Water Framework Directive [5]. It would specify more constraints in the utilization of the rivers that are utilized to produce hydroelectricity [6].

The addition of energy storage in hydropower plants can help overcome the upcoming flow regulations in rivers. In addition to this, the incorporation of an energy storage specifically in a hydropower plant can have the advantage of minimizing grid losses and transmission losses. The flexibility provided by the energy storage is not only for the hydropower, but can be translated to the entire power system. Such an addition of BESS in hydropower can help maximize the cash flows of the power plant owner while adhering to the increased water flow restrictions. As Sweden has many run-of-river hydropower plants (RoR) with and without reservoirs, the opportunity here is to implement short term regulation in RoR hydropower plants by adding BESS and not regulating the water flow in the river, to meet the environmental concerns.

1.1. Literature review

The run-of-river hydropower (RoR) contributes to a large share of the total installed hydropower capacity in Sweden. It has less flexibility to operate as they have little to no storage. It depends mainly on the water flow rate in the river and is directly influenced by seasons, thus by making the RoR act as an intermittent energy resource. As it is a synchronous source and has a spinning reserve, it possesses inertia, which contributes to grid stability. The unavailability of any stored water leads to the inflexibility to ramp the production, limiting its optimization based on market energy demand [7]. Scheduling of hydropower plants plays an important role in optimizing the production based on demand, water availability, environmental constraints and profitability from operation. Many mathematical techniques can be used to optimize a hydropower plant's operation. The existing algorithms which are commonly used to schedule hydropower plants can be observed in Table 1.

From the literature, it is observed that there is extensive research ongoing in the area of scheduling of hydropower plants using various linear and non-linear methodologies. MILP can be considered as one of the potential algorithms to schedule a hydropower plant, considering the recent advancements in the mathematical solvers and the ability to linearize functions through piece-wise linearization for reducing the computational time [21]. A significant amount of research has been done on pumped-storage hydropower plants, which provides additional flexibility in the system to optimize the overall production from the hydropower plants. Sparse literature is available on RoR scheduling techniques, with very few existing ones. Borkowski et al. (2021) employed a steady state model to analyze the effect of different hydrological conditions on a RoR hydropower plant combined with PV (Photovoltaics) and BESS. The study indicates that energy balance results are significantly influenced by flow averaging period.[22]. It is important to note that the scheduling of hydropower plants production is dependent on the electricity market prices, and it is an important parameter for unit commitment optimization of hydropower plants while they participate in the electricity market.

The electricity prices depend on various factors and there has been significant research on forecasting electricity prices in the future. The electricity prices are mainly forecasted using time series models in which the time horizons vary from hour to even weeks using historic data [23]. The existing algorithms which are commonly used to forecast electricity prices can be observed in Table 2.

All the forecasting models have their advantages and disadvantages. It is closely dependent on factors like the electricity market on which the model is employed, historical load, imports/exports, RE forecast values, historical prices, demand elasticity, bidding strategies and others [23]. There has been significant research on short- and medium-term electricity price forecasting, but studies are sparse on hourly long-term electricity price forecasting [37].

There have been batteries coupled with hydropower plants for decades to provide stability and recently such as CHBS are utilized in the regulation market [38,39]. There have been numerous studies relating to hybrid hydro-battery scheduling in the literature. Xia et al. have employed scheduling of a hybrid hydro-battery, wind, PV, thermal generator system on a multi-time scale based on operating conditions of other source of power in the system. This method is validated using the case study on New England [40]. Analysis of adding a BESS to a hydropower plant is studied by [41] to balance the net load. The study suggests that renting a battery is a better economic option rather than investing in a battery.

1.2. Contributions

Based on above literature review, there is a fundamental gap in :

- Usage of batteries and RoR hydropower plant for short-term regulation in day-ahead electricity market considering environmental restrictions for water flow
- Long-term hourly forecasting of day ahead electricity prices

With the increasing environmental concerns, the restrictions on water flow are expected to increase, which in-turn reduces the revenues which can be obtained from hydropower plants [42]. There is a potential opportunity in incorporating BESS in hydropower plants to overcome the posed environmental restrictions. This paper addresses the techno-economic feasibility of incorporating a BESS with hydropower plants and increase the flexibility in CHBS while adhering to the environmental restrictions. In addition, this paper also explores the future day-ahead electricity market prices which is critical towards profitable operation in a short term market and different scenarios are developed to address the uncertainty in future energy mix. The contributions of this paper are summarized as follows:

1. Developed a novel mixed integer linear optimization model to schedule the RoR based CHBS system with BESS for facilitating hourly short-term regulation in day-ahead market.
2. The developed model is replicable by incorporating the future water flow restrictions and future electricity prices, and scalable by using the model on any RoR based CHBS globally.
3. Developed a long-term hourly electricity price forecasting model using ANN.
4. The developed model is tested on a case study of hydropower plants in Sweden with the current water flow restrictions, and techno-economic feasibility of incorporating the BESS in hydropower plants for forecasted electricity prices until 2040.

2. Case study: Sweden

Sweden is one of the Nordic countries, with a population of 10.23 million [43]. Sweden's electricity mix is dominated by hydropower and nuclear, with a contribution of almost 40% each respectively in the electricity consumption in year 2019 [4]. Sweden's target for 2040 is to have 100% renewable electricity production and is planning to completely phase out nuclear energy production [3]. Three hydropower plants: Skattungbyn, Unnån and Hansjö which are located in the lower Oreälven river are considered in this paper. The power plants are in Dalarna County, Sweden, in between the lakes Skattungen and Orsasjön. Skattungbyn is the power plant located upstream and is a reservoir type hydropower plant. The other two hydropower plants

Table 1
Literature summary on scheduling hydropower plants.

Algorithm	Source	Findings
Mixed integer linear programming	[8]	This study included the water time delay and the result shows that the operational ability of the model is increased but makes it more computationally demanding
	[9]	The research focuses on mitigating hydro-peaking using BESS and the results indicate that the BESS will be a cost effective solution by 2025 based on the projected price of the BESS
	[10]	Scheduled hydropower plants present in Portugal considering a non-linear relationship between the water flow and net head
	[11]	Scheduled short term hydropower plants in Lancang cascade in China. The developed hierarchical model with short term hydro scheduling indicates the suggestion for the operational optimization of the power plant
	[12]	Scheduled hydropower plants considering head dependency and start up/down periods for scheduling 24 h ahead
Mixed integer quadratic programming	[13]	The proposed model has good performance considering the non-linear relationships between head, power generation and water discharge for two hydropower plants in Portugal
Non-linear approach	[14]	Schedule the hydropower plants in Portugal considering head dependency and observes that the model is not computationally demanding and has the non-linear characteristics based on implementing the model
	[15]	A non-linear programming model was designed including head dependency to schedule hydropower plants production 168 h ahead, considering three hydropower plants in Portugal
	[16]	Used non-linear programming including head dependencies to schedule 168 h ahead for seven hydropower plants in Portugal
Artificial Neural Network	[17]	Used an Artificial Neural Network (ANN) with Gaussian reduction noise algorithm for 11 hydropower plants located in Taiwan. The scheduling is done to minimize the fuel costs of thermal units and the ANN could solve the non-linearity within the optimization problem
Chaotic genetic algorithm	[18]	Scheduled the hydropower plant's production for the next 24 h while reducing premature and increased convergence speed
Stochastic modeling	[19]	Scheduled the hydropower plants in short-term to meet the requirements of bi-lateral contracts and optimize the bids to day-ahead market. The proposed method optimizes wind solar hydro complementary system to maximize the profits, meanwhile reducing the curtailment
Dynamic modeling	[20]	Used Lagrangian multipliers with dynamic modeling for 27 hydropower plants located in China to schedule based on a non-linear discrete programming.

located downstream are run of river plants with a low reservoir capacity. Hydropower is a highly treasured renewable energy resource, but it still is considered to pose some drawbacks for the local environment [44]. The operation of these three power plants is constrained by environmental restrictions, and the following are regulations to operate in Sweden:

- Minimum downstream flow rate in the plant, based on the season
- Flow rate from the upstream plant: Skattungbyn, is regulated on a weekly timeframe
- Downstream flow rate can be altered only if the flow rate has remained constant for a week

The environmental constraints have made the hydropower plants to lose their opportunity to be flexible in their electricity generation, which negatively impacts the economics of the power plant. The opportunity here is to try to revive the flexible production through the incorporation of a BESS while adhering to the environmental constraints. The flexibility can be utilized in different electricity markets, including the day-ahead and the reserve markets. In order to extract the flexibility, the technical details of the power plants and their operation conditions need to be analyzed.

The hydropower plants are connected to the local electricity grid in, SE3[1] and the technical details of the current electricity agreement from each hydropower plant is shown in Table 3. The local grid charges for these power plants are given in Table 4.

[1] Sweden is divided into 4 price areas for electricity. The four areas are: SE1—Luleå, SE2—Sundsvall, SE3—Stockholm, SE4—Malmö. Each area has a different price based on supply and demand of electricity and the transmission capacity within the regions.

Table 2
Literature summary on forecasting electricity spot prices.

Algorithm	Source	Findings
Autoregressive Moving Average (ARIMA)	[24]	Used ARIMA to predict the spot prices for 24 h ahead for Spanish and Californian markets. The average error is around 10%
	[25]	Used a combined ARIMA and GARCH model to predict spot prices for the next 24 h for Spanish market. The average error is 1%
	[26]	Used ARIMA and ANN model to predict spot prices for the next 24 h German/Austrian market. The average error is 10%
	[27]	Used seasonal ARIMA to predict the spot prices for different time horizons (24 h and monthly) for the German market. The average error is less than 11%
	[28]	Used ARIMA to predict the spot prices for 24 h ahead for the PJM interconnection in the US. The average error is 15%
Transfer function	[28]	Used transfer function to predict the spot prices for 24 h ahead for the PJM interconnection in the US. The average error is 6%
Linear regression models	[29]	Used linear regression to predict the electricity spot prices for 24 h ahead on the Spanish and Californian market. The average error is less than 5%
	[30]	Used linear regression to predict the electricity spot prices for the next 24 h on the Leipzig market. The average error is less than 10%
General Autoregressive Conditional Heteroscedastic (GARCH)	[25]	Used a combined ARIMA and GARCH model to predict the spot prices for the next 24 h on the Spanish market. The average error is 1%
	[31]	Used GARCH to predict the electricity spot prices for 24 h ahead on the Spanish and Californian market. The average error is less than 6%
Artificial Neural Networks (ANN)	[27]	Used ANN to predict the spot prices for different time horizons (24 h and monthly) for the German market. The average error is less than 10%
	[32]	Used ANN to predict the spot prices for 24 h ahead for different markets. The average error is around 10%
	[26]	Used ARIMA and ANN model to predict spot prices for the next 24 hours German/Austrian market. The average error is 10%
	[33]	Used ANN to predict the spot prices for the next 24 h for the New UK market. The average error is around 10%
	[34]	Used ANN to predict the spot prices for the short term on the US and Spanish markets. The average error is 5%
	[35]	Used ANN to predict the spot prices for Belgian market. The average error is around 10%
Support Vector Regression (SVR)	[36]	Used SVR to predict the electricity prices for different markets for different time horizons (hourly, monthly and average yearly). The average error for hourly predictions is around 10%

Based on the RoR power plant specifications, optimized operation of power plants was performed until 2040 through a two-stage algorithm as discussed in Section 3.

3. Methodology

The methodology on the paper is divided in two models:

- Future Electricity Price Forecasting Model (FEPFM)
- Battery Profitability Model (BPM)

The FEPFM deals with the forecasting of electricity prices for different scenarios to evaluate the performance and optimal scheduling to maximize profits yielded by the battery energy storage in the future scenarios using the BPM. The BPM models the battery energy storage system operating together with the hydropower plant and determines the optimal profits that can be achieved given the technical, economic, and environmental constraints. The models are softly linked with one another, i.e., the results from FEPFM are used in BPM. Both the models are developed using Python.

3.1. Future electricity price forecasting model (FEPFM)

The future electricity prices are used as an input for the BPM to optimize the operation and analyze the performance of the CHBS in the future. The prediction of electricity prices can be based on numerous mathematical models and various machine learning algorithms. Several machine learning models were tested, and in FEPFM the following machine learning methods are considered which were able to give satisfactory performance.

- Original Least Square Regression (OLSR)
- Lasso Regression (LR)
- Ridge Regression (RR)
- Support Vector Regression (SVR)
- Artificial Neural Network (ANN)

The algorithms are analyzed in FEPFM to identify the best-fit algorithm and use it for predicting the future electricity prices from 2020 to the year 2040. The steps which are used to implement the algorithms are data selection, data splitting, training the model, testing the model, and predicting the future values.

3.1.1. Scenarios

To account for the uncertainties in the future, three different scenarios were developed based on Sweden's nuclear energy targets for 2040 as shown in Table 5.

The electricity production from wind power, solar power and Combined Heat and Power (CHP) are increased over the years from 2020 to 2040 to meet the increasing demands and to fill the void created by removing nuclear power plants in scenarios 1 and 2. The load is projected for each scenario from 2020 until 2040 and the values are shown in Table 6.

In scenario 1, 100% RE production is expected by 2040 and the contribution from each energy source can be seen in Table 6 following a report from Svensk Vindenergi, which predicts the energy production in the year 2040 [45]. In scenario 2, two nuclear power plants were still in operation by 2040 and for scenario 3, the nuclear production remains the same from 2020 until 2040. The production from wind energy and CHP were considered to increase linearly to meet the target production in the year 2040 and for nuclear energy, power plants have been decommissioned in different years to meet the nuclear targets for different scenarios accordingly as shown in Table 8.

Table 3
Specifications of the hydropower plants.

	Skattungbyn	Unnån	Hansjö
Head (m)	4.8 – 7.5	6.5	10.6
Turbine type	Kaplan	Kaplan	Kaplan
Number of turbines	2 (0.7 and 0.7 MVA)	2 (1.4 and 1.4 MVA)	1 (3.3 MVA)
Maximum electricity which can be supplied from the grid (MW)	1.4	2.9	3.3
Maximum electricity which can be consumed from the grid (MW)	0.1	0.06	0.07

Table 4
Grid agreement costs [76].

	Charges	Units
Fixed electricity network charge	8 50	€/ year
Electrical measurement charge	4 50	€/ year
Annual power charge	3.6	€/ kW/year
Transmission line charge	0.074	€/ kWh/km

Table 5
Scenario summary.

Scenario 1	A scenario in which nuclear production is completely eradicated from the Swedish electricity mix by the year 2040
Scenario 2	This scenario is an alteration of scenario 1 in which nuclear production is decreasing over the years, but has not been completely eradicated. More details are provided in Table 6.
Scenario 3	This scenario is based on the installed nuclear capacity present in 2020 and this installed capacity will remain the same until 2040

Table 6
Electricity production in 2040.

Energy sources	Electricity production in 2040		
	Scenario 1	Scenario 2	Scenario 3
Wind	90 TWh	82 TWh	60 TWh
Hydro	70 TWh	70 TWh	70 TWh
CHP	38 TWh	33 TWh	27 TWh
Nuclear	0 TWh	18 TWh	48 TWh
Solar	10 TWh	5 TWh	3 TWh
Total production	208 TWh	208 TWh	208 TWh

3.1.2. Data

The machine learning model uses historical data to form relationships between the input and output variables. The historical data were accessed from Svenska kraftnät [46] and Nordpool [47] from 2013 to 2019. The output for the FEPFM is the day ahead wholesale electricity price and to identify the inputs for FEPFM, different variables were analyzed. Based on the performance of the model, the selected inputs are described in Table 7

Different combinations of input data set with lagged variables were tested in FEPFM. The input data for the model used begins with 35 input variables without using the lagged price, and this is treated as input data 1. Then the input data is being revised to identify the best combination of historical time-lagged variables which yields the best accuracy of the model. There have been four such variations of the time-lagged variable which are considered in the model as given in Table 9. These four variables have been selected based on previous research articles [23,24,28] and the input dataset, which yields higher accuracy and most significant changes. If the price of electricity at the time 't' for day 'd' is to be forecasted, then the lagged time variables considered for different input data in the model are shown in Table 9, where *'i'* is the specific hour considered at day *'d'*, P_t is the price of electricity in €/MWh at the time *'t'* for *'d'*, $P_{avg,i}$ is the average price of electricity at day *'d-i'*.

3.1.3. Future input variables prediction

The input variables considered for FEPFM are predicted for the future to be able to forecast the electricity prices. These variables are predicted based on the historical data with future estimates and assumptions. The exact method for obtaining the future values of the parameters is explained in the appendix. Once the future profile for each input variable is identified, it is fed into FEPFM to predict the electricity prices. The FEPFM model is soft linked to BPM model, which is discussed in detail in the following Section 3.2.

3.2. Battery profitability model (BPM)

The operations of the hydropower plant operating together with the battery energy storage system is modeled as a mixed-integer linear programming algorithm. The model is a rolling horizon optimization model which aligns the horizon to the day ahead electricity market and the model optimizes the battery for the day ahead from hour 13 (as per the operation of Swedish electricity market) the previous day. Hence, a schedule for 35 h is obtained. The model, when provided with hourly water flow in the river and hourly electricity market prices, optimizes the operation of the plant with an objective of maximization of the revenue. The model is further divided into two interdependent parts: Turbine Model and Battery Model.

The turbine model is designed to model the water flow through the turbine and the spillways while computing the corresponding power production. The battery model deals with the charging, discharging and storage level in the battery. It monitors the depth of discharge of the BESS, degradation of the BESS, and costs incurred related to it. BESS is connected to the bus bar where the hydropower plant is connected to the substation. The electricity generated in the power plant is used to charge the battery, and the battery discharges to supply electricity to the grid. This system is simulated from the year 2020 until the year 2040. The life of the battery is determined by the capacity degradation of the battery. The capacity degradation can be classified into two major types considered in the model: calendar degradation and cycling degradation, based on its operation. Once the battery is degraded it is then replaced by a new battery and this happens until the end of the simulation period. The algorithm which is used in BPM is a mixed-integer linear programming model which has the objective of maximizing the revenue from the CHBS. The parameters and decision variables are discussed in the upcoming sections.

3.2.1. Parameters

Parameters have a predefined value and cannot be changed by the model. The BPM will be using the following variables as indices: t ∀ 1:24 time in hours in one day j ∀ segment in Power vs Flow Curve k ∀ segment in cost incurred for a charge discharge cycle vs depth of discharge

The parameters considered in BPM can be viewed in Table 10 and the fixed values of parameters which are not dependent on time or water flow can be observed in Table 11.

3.2.2. Decision variables

These are the variables which are subjected to vary and optimized by the model to choose the best value to get the bet result. The decision variables used in this model can be observed in Table 12.

Table 7
Inputs for FEPFM.

Inputs selected	Description
Hour of the day	Hours of the day are considered from 1 to 24.
Day	Divided into workdays and weekends. Workdays are from Monday to Friday and weekends are Saturday and Sunday.
Season	Sweden experiences four seasons: Summer, Autumn, Winter and Spring.
Load in Sweden (MWh/h)	It is the consumption of electrical energy in Sweden in a specific hour.
Hydro-reservoir amount (MWh)	It is the amount of hydropower potential available with the water resources in Sweden. It is calculated based weekly.
Wind energy production (MWh/h)	It is the amount of electrical energy produced by the wind power plants in Sweden in a specific hour.
CHP energy production (MWh/h)	It is the amount of electrical energy produced by the CHP power plants in Sweden in a specific hour.
Nuclear energy production (MWh/h)	It is the amount of electrical energy produced by the Nuclear power plants in Sweden in a specific hour.
Lagged electricity price (€)	It is the price of electricity in the previous hours of the specific considered hour.
Lagged average electricity price of the previous days (€)	It is the average price of electricity of the previous day of the specific considered day.

Table 8
Decommissioning of nuclear power plants.

Reactor name	Reactor capacity (MWe)	Scenario 1		Scenario 2		Scenario 3	
		Year of decommissioning	Nuclear capacity in grid after decommissioning [MWe]	Year of decommissioning	Nuclear capacity in grid after decommissioning [MWe]	Year of decommissioning	Nuclear capacity in grid after decommissioning [MWe]
Ringhals 1	881	2020	6 872	2020	6 872	2020	6 872
Forsmark 1	990	2023	5 882	2024	5 882	–	6 872
Forsmark 2	1 118	2026	4 764	2028	4 764	–	6 872
Ringhals 3	1 062	2029	3 702	2032	3 702	–	6 872
Ringhals 4	1 130	2033	2 572	2036	2 572	–	6 872
Forsmark 3	1 172	2037	1 400	–	2 572	–	6 872
Oskarshamn 3	1 400	2040	0	–	2 572	–	6 872

Table 9
Time lagged variables for different input datasets.

Input data	Time lagged variables considered
Input data 1	–
Input data 2	$P_{avg,1}$
Input data 3	$P_{avg,1}$,$P_{avg,2}$
Input data 4	P_{t-24}, $P_{avg,1}$, $P_{avg,2}$
Input data 5	P_{t-24}, P_{t-48}, $P_{avg,1}$, $P_{avg,2}$

3.2.3. Objective function

The objective function defined in Eq. (1) is to maximize the revenue of the model. Revenue can be increased by increasing the cash inflow and decreasing the expenses. The cash inflow is increased by increasing the power supplied to the grid at high price hours, which is given by $P_{G,t} * \lambda_t$, where $P_{G,t}$ is the power fed into the grid at time t and the expenses are reduced by decreasing the cost incurred, which is given by $-N_t$.

$$Maximize \sum (P_{G,t} * \lambda_t - N_t) \tag{1}$$

3.2.4. Constraints

Constraints are the equations that limit the decision variable values and act as a boundary limit. The constraints for turbine modeling are as follows:

$$-P_{G,t} + P_{T,t} - CS_t + DS_t = 0; \forall t \tag{2}$$

The hourly electrical power balance in the system is given by Eq. (2). Power enters the system from the turbine $P_{T,t}$ and battery discharges at DS_t, and then the power exits the system by feeding electricity into the grid $P_{G,t}$ and charging of the battery CS_t.

$$P_{G,t} \geq 0 \tag{3}$$

$$P_{G,t} \leq \overline{P_G} \tag{4}$$

The power fed into the grid $P_{G,t}$ is constrained as a non-negative variable in Eq. (3) as the power is only supplied to the grid and not bought from the grid. The upper bound is defined in Eq. (4) which is the maximum power $\overline{P_G}$ that can be fed into the grid.

$$Q_{T,t} + Q_{S,t} = Q_t; \forall t \tag{5}$$

Conservation of mass flow in the system is given by Eq. (5). The mass flow enters the system by the flow in the river upstream Q_t and is equal to the sum of turbine $Q_{T,t}$ and the spillway $Q_{S,t}$. Due to the regulations from environmental constraints, the hydropower plant has to maintain the flow in the river. When the water inflow is above the turbine's limit, the water is diverted through the gates acting as spillway and not stored in the reservoir. In this way the plant operator maintains the same flow in the downstream and the upstream of the river.

$$Q_{T,t} \geq 0 \tag{6}$$

$$Q_{S,t} \geq 0 \tag{7}$$

The flow through turbine $Q_{T,t}$ and spillway $Q_{S,t}$ have minimum and maximum bound defined in Eqs. (6) and (7).

$$Q_{T,t} \leq \overline{Q_T} \tag{8}$$

Table 10
Parameters used in BPM.

Symbol	Units	Description
Turbine model		
λ_t	€	Day ahead electricity price for SE3 region at time 't'
Q_t	m^3/s	Flow in the river at time 't'
$\overline{Q_s}$	m^3/s	Maximum flow of water to the spillway
$\overline{Q_T}$	m^3/s	Maximum flow of water to the turbine
$\overline{P_G}$	MW	Maximum power generation in turbine
$\overline{Q_j}$	m^3/s	Minimum Water Flow at segment 'j'
$\overline{P_j}$	MW	Minimum Power for a segment 'j'
$\overline{Qx_j} = \overline{Q_{j+1}} - \overline{Q_j}$	m^3/s	Segment 'j' step size
$m_j = (\overline{P_{j+1}} - \overline{P_j})/(\overline{Q_{j+1}} - \overline{Q_j})$		Slope of Power vs Flow for segment 'j' and ' j+1'
Battery model		
$\overline{S}$	MWh	Battery Capacity
$\overline{CS}$	MW	Maximum charging power for the battery
$\overline{DS}$	MW	Maximum discharging power for the battery
η_{CS}	%	Efficiency of charging of the battery
η_{DS}	%	Efficiency of discharging of the battery
$\overline{N_k}$	€	Cost incurred during cycle degradation for avs specific DOD segment 'k'
$\overline{DOD_k}$		Step change in depth of discharge in cost incurred for a charge–discharge cycle vs depth of discharge at segment 'k'
$m_j = (\overline{N_{k+1}} - \overline{N_k})/(\overline{DOD_k})$	€	Slope of the cost incurred for a charge–discharge vs depth of discharge graph for segment 'k'

$$Q_{S,t} \le \overline{Q_S} \tag{9}$$

The maximum permissible flow in the turbine and spillway is defined in the Eqs. (8) and (9).

$$Q_{T,t} = \sum_j (\overline{Q_j} * z_{j,t} + Q_{T,j,t})); \forall t \tag{10}$$

To optimize the power production from the water flow in the river, the piecewise linear function is used to determine the turbine operation point with known operation data points. Eq. (10) determines in which segment the operating point lies, in the piece wise linear function.

$$P_{T,t} = \sum_j (\overline{P_j} * z_{j,t} + m_j * Q_{T,j,t})); \forall t \tag{11}$$

Eq. (11) determines the power output for the operating flow point in the turbine.

$$\sum_j z_{j,t} = 1; \forall t \tag{12}$$

Table 11
Fixed values of parameters in BPM.

Parameter	Value (Skattungbyn/Unnån/Hansjö)	Units
$\overline{Q_s}$	220/200/200	m^3/s
$\overline{Q_T}$	24/46/45	m^3/s
$\overline{P_G}$	1.17/2.8/3.6	MW
$\overline{CS}$	$\overline{S} * Crating$	MW
$\overline{DS}$	$\overline{S} * Crating$	MW
η_{CS}	98	%
η_{DS}	98	%
S_t	$0.2 * \overline{S} \le S_t \le 0.95 * \overline{S}$	%

Table 12
Decision variables used in BPM.

Symbol	Units	Description
Turbine model		
$Q_{T,t}$	m^3/s	Flow of water through the turbine 'T' at time 't'
$Q_{S,t}$	m^3/s	Flow of water through the spillway at time 't'
$Q_{T,j,t}$	m^3/s	Flow of water through the turbine at time 't' for segment 'j'
$P_{G,t}$	MW	Power supplied to the grid at time 't'
$P_{T,t}$	MW	Power produced in the turbine 'T' at time 't'
$Z_{j,t}$		Binary variable for piecewise linear power flow curve in segment 'j' at time 't'
Battery model		
S_t	MWh	Storage in the battery at end of time 't'
CS_t	MW	Power spent in charging the battery in time 't'
DS_t	MW	Power spent in discharging the battery in time 't'
$\mathbf{u_{CS,t}}$		Charge state variable at time 't'
$\mathbf{u_{DS,t}}$		Discharge state variable at time 't'
$D_{tot,t}$	MWh	Total discharge energy at end of time 't' for one cycle
DOD_t	MWh	Depth of discharge at end of cycle at time 't'
$DOD_{j,t}$	MWh	Depth of discharge at time 't' for segment 'j'
N_t	€	Cost of degradation of the battery for a cycle in time 't'
$Z_{DOD,k,t}$		Binary Variable for piecewise linear curve of the cost incurred for a charge–discharge cycle vs depth of discharge in segment 'k' at time 't'

$$Q_{T,j,t} \le z_{j,t} * \overline{Qx_j}; \forall t, \forall j \tag{13}$$

Eqs. (12) and (13) are used to implement a piecewise linear function.

The constraints for battery modeling are as follows:

$$S_t = S_{t-1} + (\eta_{CS} * CS_t) - (\eta_{DS} * DS_t); \forall t > 1 \tag{14}$$

The battery storage's hourly energy conservation is defined in Eq. (14). The previous hour's remaining storage S_{t-1} is carried forward to the current hour S_t. Charging of the battery CS_t is an input to the system and the discharging of the battery DS_t is the output of the system. The charging and discharging efficiencies of the battery system are included in the equation.

$$CS_t \le u_{CS,t} * \overline{CS}; \forall t \tag{15}$$

$$CS_t \geq 0; \forall t \tag{16}$$

$$DS_t \leq u_{DS,t} * \overline{DS}; \forall t \tag{17}$$

$$DS_t \geq 0; \forall t \tag{18}$$

Eqs. (15), (16), (17) and (18) provide the constraints for the lower bound and upper bound of the charging rate CS_t and discharging rate DS_t of the battery. The power spent at charging and discharging the BESS is limited by the charge and discharge state variable and the maximum charging and discharging power for the BESS.

$$S_t \leq 0.95 * \overline{S} \tag{19}$$

$$S_t \geq 0.2 * \overline{S} \tag{20}$$

The lower bound and upper bound of the decision variable storage level is defined in the Eqs. (19) and (20). The lower operating SoC of the battery is considered as 20% based on this study [48] which states that the battery cycled at SoC range 20%–80% has lower degradation compared to operating beyond these limits. The upper operating limit for the battery is considered as 95% based on the charging curve of the battery. The battery can be charged at 1C only till 95%, after that the battery has to be charged at a lower current which is not considered in this study.

$$u_{CS,t} + u_{DS,t} = 1; \forall t; u \in 0, 1 \tag{21}$$

The state of the battery can be binary, charging $u_{CS,t}$ or discharging $u_{DS,t}$, which is defined by Eq. (21).

$$D_{tot,t} = D_{tot,t-1} + DS_t - DOD_t; \forall t > 1 \tag{22}$$

$$DOD_t = \overline{S} * \sum_k (\overline{DOD_k} * Z_{DOD,k,t} + DOD_{k,t}); \forall t \tag{23}$$

$$N_t = \sum_k (\overline{N_k} * Z_{DOD,k,t} + DOD_{j,t} * m_{N,k}); \forall t \tag{24}$$

$$\sum_k Z_{DOD,k,t} \leq 1; \forall t \tag{25}$$

$$DOD_{k,t} \leq Z_{DOD,k,t} * \overline{DOD_k}; \forall t, \forall k \tag{26}$$

$$0 \leq D_{tot,t} \leq u_{DS,t} * \overline{S}; \forall t \tag{27}$$

$$0 \leq DOD_t \leq u_{CS,t} * \overline{S}; \forall t \tag{28}$$

$D_{tot,t}$ is a cumulative sum of hourly discharge of a discharge cycle ($u_{CS,t} = 0$ and $u_{DS,t} = 1$) as shown in Eq. (22) where DOD_t is forced to zero during the discharge cycle, which is given by Eq. (28). When the BESS switches from discharging state to charging state ($u_{CS,t} = 1$ and $u_{DS,t} = 0$), $D_{tot,t}$ is forced to zero by Eq. (27). The total discharge of a cycle is stored as the last cumulative sum of discharge $D_{tot,t-1}$ (at the previous hour) which is assigned to the variable DOD_t as shown in Eq. (22). With the total discharge of a discharge cycle DOD_t, the corresponding cost for the cycle due to degradation is calculated using piecewise linearization (further explained in Appendix B) in Eqs. (23) to (26). This cost per cycle N_t is part of the objective function shown in Eq. (1) where the cost is minimized.

With the operation of BPM explained, the evaluation of the BPM would be based on the revenue obtained from CHBS and can be explained by the following economic parameters.

3.2.5. Economic parameters

Once the production of the combined hydropower and battery system is optimized, based on the predicted future electricity prices, the revenue from the operation of the combined system is calculated. To compare various cases in this research, Net Present Value (NPV) is used as a measurement tool. NPV is one of the methods to identify the profitability of the implemented BESS in the hydropower plants by making projections for the future while considering the discounted cash flows and the initial investment cost.

The formula for NPV is as shown below:

$$NPV = \sum_{i=1}^{n} \frac{Cashflow_i}{(1+r)^i} - InitialInvestment \tag{29}$$

Where i is the year, n is the total years of operation i.e., 20 years, r is the discount rate.

3.2.6. Inputs and assumptions for BPM

A total of, 11016 cases were formulated in BPM based on varying the following factors:

- Electricity Price: three different future electricity price scenarios were obtained from FEPFM.
- Water Flow: Three different flow scenarios were developed based on the operational regulations, which follow the environmental restrictions in the river and the historical inflow pattern. The environmental constraints consider the livelihood of flora and fauna in the river, so, the effect on them by the operation of hydropower plants in the river, is minimalistic. Due to this, the flow scenarios have a high flow during the spring flood time (beginning of spring). In Fig. 1 the flow at the Skattungbyn reservoir and in Fig. 2 the flow at Unnån and Hansjö which includes the flow from tributaries are defined as three scenarios: low flow, medium flow and high flow based on assumptions from the wettest year and the driest year over the past 30 years.
- Battery Connection Point: The number of BESS installations is dependent on the connection point of the hydropower plant to the grid. It can be installed in two locations as Skattungbyn has a different connection point to the grid, Unnån and Hansjö are connected to the same point.
- Battery Size: Six different battery sizes based on plant size were selected based on the RoR capacity, existing grid limitations and the costs to extend the current grid limitations as shown in Table 13. These selected battery sizes were then used to evaluate the performance of CHBS.
- Grid Agreement Extension : Considering the current grid agreement, a case was developed for an increased capacity grid agreement to analyze the effect of higher power injection into the grid on revenues. A case with a 1 MW increased capacity grid agreement is considered on top of the existing limit of 1.4 MW for Skattungbyn, 2.9 MW for Unnån and 3.3 MW for Hansjö.
- Battery Cost: the cost of the battery was varied from 100% to 0% of its current cost in steps of 2% to calculate the break-even point of battery costs.

The methodology of BPM is illustrated in Fig. 3

Following parameters were fixed and assumed to be constant in the BPM:

- The BPM is evaluated for 2020 to 2040 with the electricity price obtained from FEPFM.
- The discount rate is considered as 6%.
- The c rating of the BESS is assumed to be 1C.
- After the end of the life of a battery which is 60% capacity of initial capacity [42], it is replaced by a new battery.
- The capacity extension of the grid agreement costs 3.6 € / kW/year from Table 4.

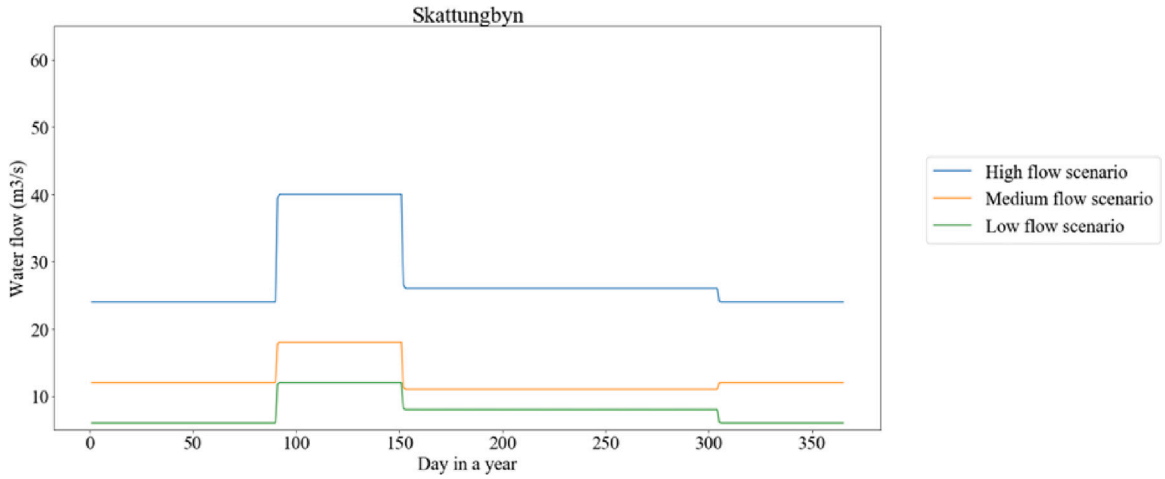


Fig. 1. Skattungbyn flow.

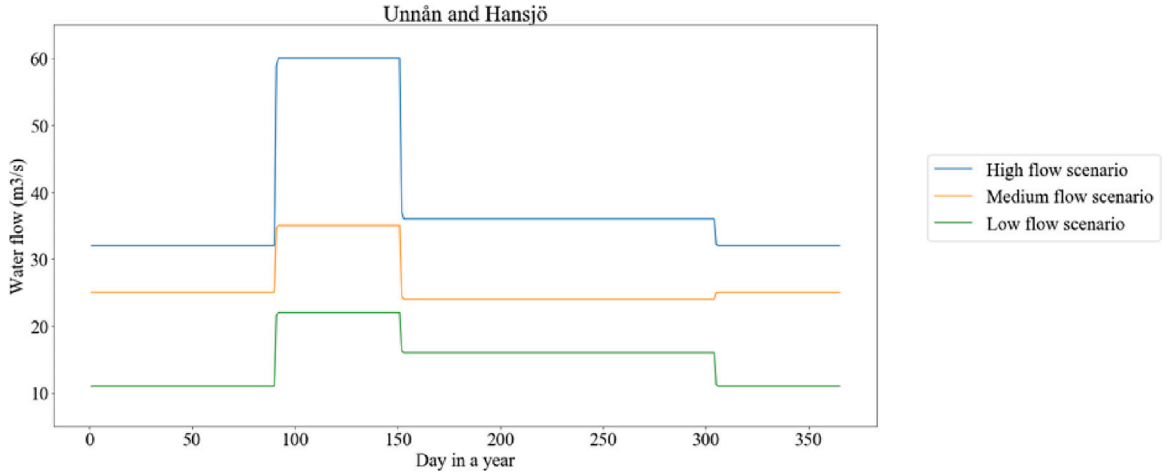


Fig. 2. Unnån and Hansjö flow.

Table 13
Battery sizes considered.

Connection point	Battery size (MWh)
Skattungbyn	0.75
	1.00
	1.25
	1.50
	1.75
	2
Unnån and Hansjö	2.75
	3.25
	3.75
	4.5
	5
	5.5

- Utility-scale BESS costs were obtained from National Renewable Energy Laboratory (NREL) which includes power electronics and project development costs. The BESS costs 364 €/kWh. The battery alone amounts to 55% of the BESS costs, and the cost of power electronics amounts to 163.8 € / kWh [49]. On evaluating the existing power system at the hydropower plants, the BESS installation needs a separate transformer as the existing transformer is limited by its capacity and operating voltage, which is different from the BESS's inverter output voltage. In a BESS, an inverter comes integrated to the battery. Therefore, the costs of both the inverter and transformer have been considered in the BESS costs from [49].
- The salvage costs of the batteries has not been considered to calculate the NPV due to unknown values. The recycling of Li-ion batteries are currently low, but by 2040 the recycling industry for Li-ion batteries would completely be different due to technological advancements.
- As the FEPFM is performed for long term predicting, there is a possibility that the data drift might be encountered. This can be corrected by training the model at regular time intervals when the actual data is known. But due to lack of actual data, this has been skipped in this study.

4. Results and discussion

The results from FEPFM and BPM are discussed in detail in this section.

4.1. FEPFM

The results of using different models and input data are shown in Table 14. The data was split into testing (30%) and training test (70%) and the performance was measured using Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE) and Mean Percentage Error (MPE).

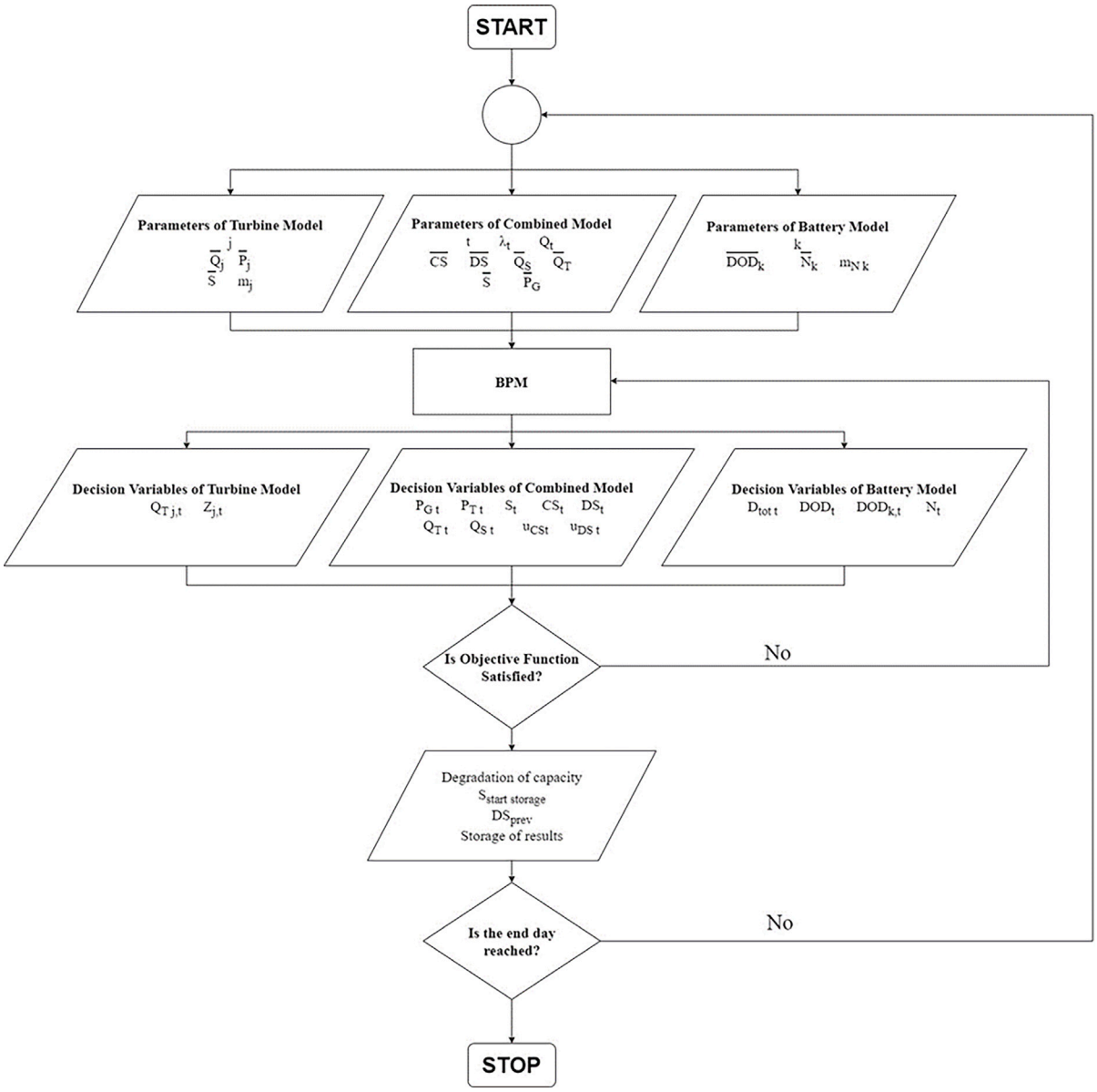


Fig. 3. Schematic explanation of the process employed in BPM.

From the above tables, it is observed that the value of errors has been significantly decreasing as a time-lagged variable is included in the input data, i.e., from input data 1 to input data 2. Hence, it can be said that the time-lagged variables affect the model accuracy. Also, there has been a significant decrease in errors when the average price of two previous days are considered, i.e., from input data 3 to input data 4. All the models perform better when the prices before 24 and 48 h are considered, i.e., from input data 4 to input data 5, except for neural network; it performs better only when the prices before 24 h were considered i.e., input data 4. It can also be seen that the accuracy of regression is almost the same in all the models with the same input, but the lowest value of MAPE and MPE are obtained using Artificial Neural Network (ANN) for the input dataset 4. Hence, ANN was considered for the paper for predicting the electricity prices in different scenarios. Similar to the literature, FEPFM has high accuracy, and it is in par with the existing literatures. A sensitivity analysis was performed by varying the inputs individually from −75% to +75% to check the influence of inputs on the electricity price. The input data for the year 2020 in Scenario 1 was used in this analysis and the results are shown in Fig. 4.

From the above sensitivity analysis, it is observed that the load and CHP production are directly related to the price. This can be explained by two reasons:

- CHP production is like that of the load. It increases during winter and decreases during summer and a similar equation fit for CHP production is calculated to that of the load
- The prices of electricity have been low in summer where CHP production is low and high in winter where CHP production is high due to the fact that the CHP is operated in Sweden based on the heating demands. Typically, the prices are high in winter when the CHP production is high and the prices are low in summer when the CHP production is low.

On the other hand, nuclear, wind production and hydro reservoir amount are indirectly proportional to the electricity price, if it increases, the electricity price decreases. The most significant influence corresponds to load and wind energy production. A 75% increase in load, increases the average electricity price by 110% and for a 75% decrease in wind energy production, the average electricity price is increased by 100%. In addition to that, it is important to note that the

Table 14
Results for different inputs using different machine learning algorithms.

Algorithm used	Input data	MAPE (%)	MPE (%)	MSE	RMSE
OLS Regression	Input data 1	32.17	−10.33	15984.2	126.43
	Input data 2	15.57	−4.1	4181.86	64.75
	Input data 3	15.38	−3.96	4033.41	63.51
	Input data 4	14.24	−3.54	3700.17	60.83
	Input data 5	14.13	−3.47	3702.36	60.84
Ridge Regression	Input data 1	32.18	−10.35	15992.4	126.46
	Input data 2	15.58	−4.1	4175.53	64.62
	Input data 3	15.36	−3.95	4025.88	63.45
	Input data 4	14.23	−3.55	3695.71	60.79
	Input data 5	14.12	−3.47	3696.74	60.8
Lasso Regression	Input data 1	32.17	−10.34	15985.5	126.43
	Input data 2	15.56	−4.1	4178.4	64.64
	Input data 3	15.34	−3.96	4028.97	63.47
	Input data 4	14.22	−3.54	3695.72	60.79
	Input data 5	14.1	−3.47	3698.3	60.81
Support Vector Regression	Input data 1	32.61	−13.39	15510.1	124.54
	Input data 2	15.55	−5.74	4192.36	64.75
	Input data 3	15.235	−5.455	23996.7	63.22
	Input data 4	13.77	−4.211	3634.88	60.3
	Input data 5	13.68	−4.11	3625.37	60.21
Artificial Neural Network	Input data 1	28	−8	12868.4	113.43
	Input data 2	14.7	−3.6	3913.85	62.56
	Input data 3	14.1	−3.4	3535.37	59.45
	Input data 4	13.0486	−2.95	3254.94	57.05
	Input data 5	13.31	−2.98	3357.45	57.9

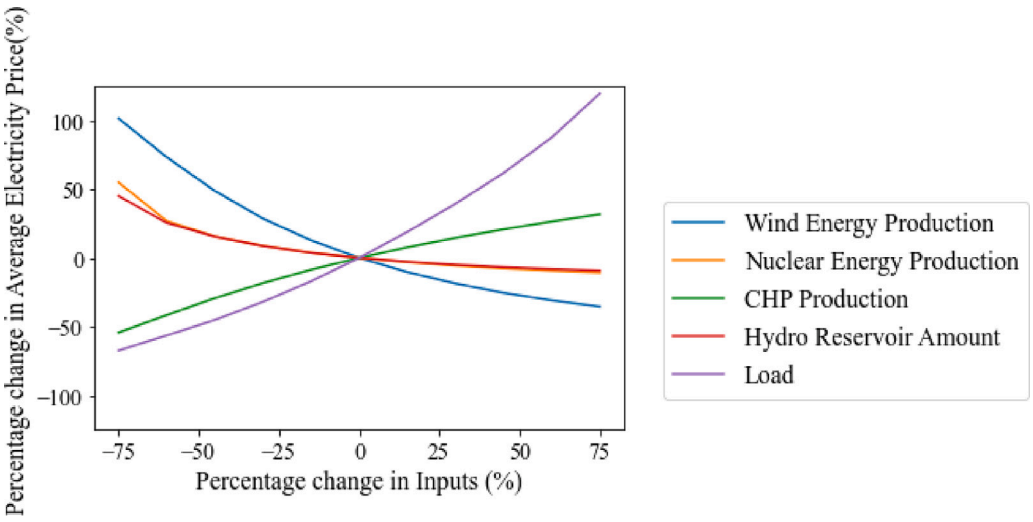


Fig. 4. Sensitivity analysis of inputs.

production of wind energy depends on wind availability and similarly hydro reservoir amount depends on the precipitation. This sensitivity analysis was performed to analyze the influence of different parameters on the electricity price, and it is not used to compare among the parameters. The variation of electricity prices throughout the year, in years 2020 and 2040 for all three scenarios are shown in Fig. 5. Results from the three scenarios indicate that the trend of electricity prices has been raising, and one major contribution to this is from the increasing load. The lower limit of the prices has been the same over the years, occurring in the summer months when the load is minimum and hence there is no requirement for heating. The prices of electricity have been relatively higher at the beginning of the year than during the summer months. The electricity prices have been the highest during February and March for all the scenarios considered. The high prices in February and March can be explained by the decreasing nuclear production and decreasing hydro reservoir amount. This can be attributed to disregarding the possible flexible technologies in the future along with possible large scale storage solutions. The low prices in summer are due to lower demand. Scenario 1 is observed to have the highest price volatility. There is a need for another source of energy, either CHP or energy storage technology, to fill up the void left by nuclear in the base load. Scenario 2 seems to be an achievable and realistic scenario in which nuclear energy is not completely phased out but still is in the energy mix and provides stability to the system. Scenario 3 is observed to have the least price volatility due to the presence of a larger share of nuclear energy in the energy mix. Hence,

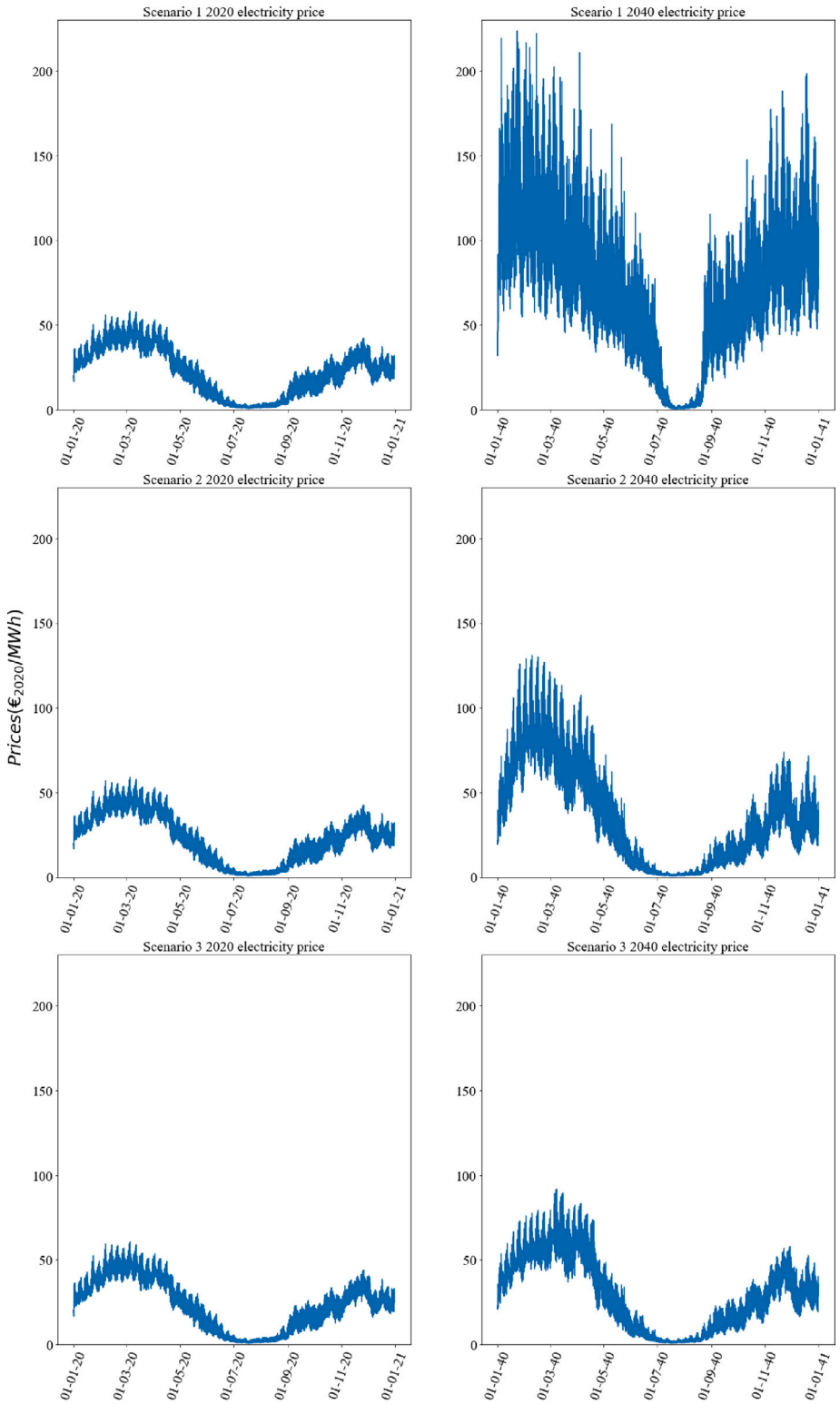


Fig. 5. Forecasted electricity prices from FEPFM.

the prices in scenario 3 are the most stable and scenario 1 is the most volatile. The results of predicted electricity prices in different scenarios are used in the BPM to assess the profitability of the BESS until 2040.

With the electricity prices forecasted until 2040 from FEPFM, the BPM can be utilized to simulate the optimization of the CHBS. The results from BPM are shown below.

4.2. BPM

To explain the working of the BPM, a typical case is studied as follows: The prices from scenario 1 and river flow from the medium flow scenario were selected along with a 5 MWh battery in Unnån and Hansjö to explain the working of the scheduling of CHBS in BPM during the winter. The week starts from Monday and ends with Sunday, and the production planning of the CHBS can be observed in Fig. 6.

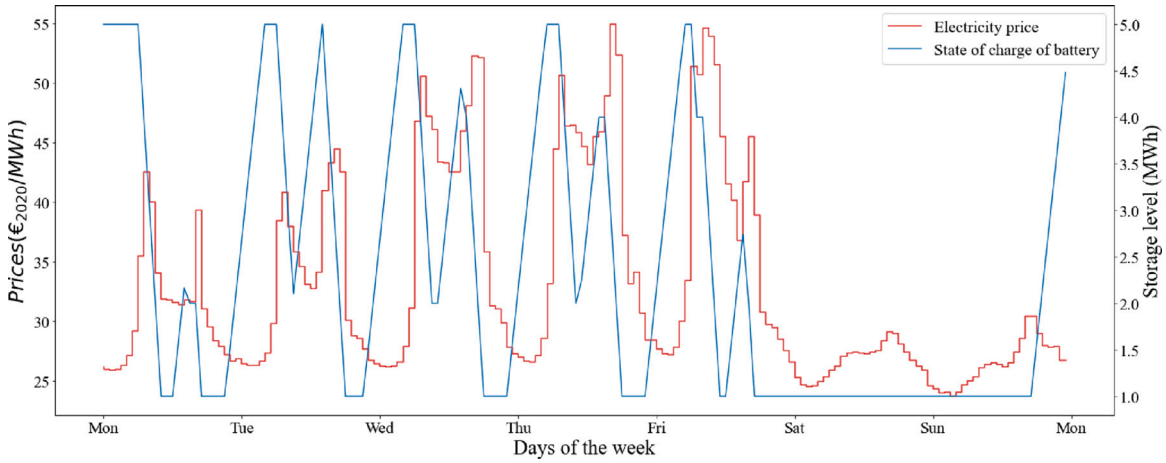


Fig. 6. Production planning of the CHBS.

Table 15
Influence of different price scenarios in Skattungbyn.

Battery size (MWh)	Grid agreement extension	Water flow	Scenario	Discounted cashflows over duration (€)
1	No	Medium	3	31433
1	No	Medium	2	41378
1	No	Medium	1	50639
1.75	Yes	Medium	3	11609
1.75	Yes	Medium	2	25887
1.75	Yes	Medium	1	44116

Table 16
Influence of different price scenarios in Unnån and Hansjö.

Battery size (MWh)	Grid agreement extension	Water flow	Scenario	Discounted cashflows over duration (€)
3.25	No	Medium	3	102589
3.25	No	Medium	2	134758
3.25	No	Medium	1	165804
5	Yes	Medium	3	73784
5	Yes	Medium	2	124179
5	Yes	Medium	1	172071

In Fig. 6, the BESS charges when the price of electricity is low, which is reflected in the BESS state of charge curve in the very same figure. BESS discharges to the grid and decreases the storage level when the price of electricity is high, i.e., the battery discharges when the prices are relatively high, thus generating revenue. During the weekend, the prices of electricity are relatively low compared to the workdays and the volatility of the electricity price is also low. Hence, the operation of the battery is restricted since the profits yielded by operating the battery does not cover the levelized cost per cycle of the battery. During the weekdays, the electricity prices have clear high peaks and the battery charges and discharges throughout the workdays based on the electricity price as observed in Fig. 6. With this operation of the battery, various cases based on varying battery size, battery connection points, alternate grid agreements and future electricity prices were simulated, and the techno-economic feasibility of the system considering environmental constraints was analyzed.

4.3. Economic analysis

The economic analysis uses NPV to evaluate the different cases. Due to the presence of two different battery connection points, one battery in Skattungbyn and a separate battery for Hansjö and Unnån together were analyzed. In order to capture the influence of different price scenarios, for a fixed battery size and a flow scenario, the cash flows from the CHBS is shown in Table 15.

From Table 15, it can be observed that when the price scenario is changed from 3 to 1, the discounted cashflow increases. The reasoning behind this would be that scenario 1 has a high volatile prices when compared to that of scenario 3. The similar results can be observed for the CHBS in Unnån and Hansjö. The results for the CHBS placement in Unnån and Hansjö can be observed in Table 16.

The influence of different scenarios had a direct effect on the overall discounted cashflows. The cases were analyzed having the medium water flow for the overall time horizon considering a specific battery size. The effect of different flow scenarios can be observed in Tables 17 and 18

Table 17
Influence of different flow scenarios in Skattungbyn.

Battery size (MWh)	Grid agreement extension	Scenario	Water flow	Discounted cashflows over duration (€)
1	No	2	Low	40474
1	No	2	Medium	41378
1	No	2	High	41597
1.75	Yes	2	Low	23598
1.75	Yes	2	Medium	25887
1.75	Yes	2	High	26344

Table 18
Influence of different flow scenarios in Unnån and Hansjö.

Battery size (MWh)	Grid agreement extension	Scenario	Water flow	Discounted cashflows over duration (€)
3.25	No	2	Low	134864
3.25	No	2	Medium	135758
3.25	No	2	High	135924
5	Yes	2	Low	118892
5	Yes	2	Medium	124179
5	Yes	2	High	125767

The flow rate plays a significant role in the CHBS. The effect of this flow in Skattungbyn can be seen in Table 17. The table shows the discounted cash flow in the CHBS with Scenario 2 as the electricity prices and shows the effect of flow rate based on a fixed battery size for different grid agreement settings. Low flow scenario has the least cash flow, which can be explained by the limited availability of power

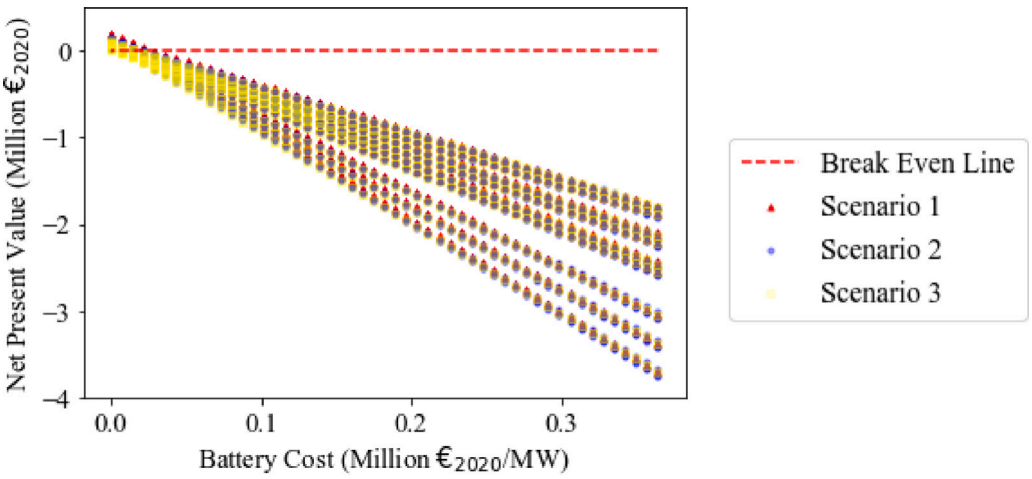


Fig. 7. Results from BPM for Skattungbyn.

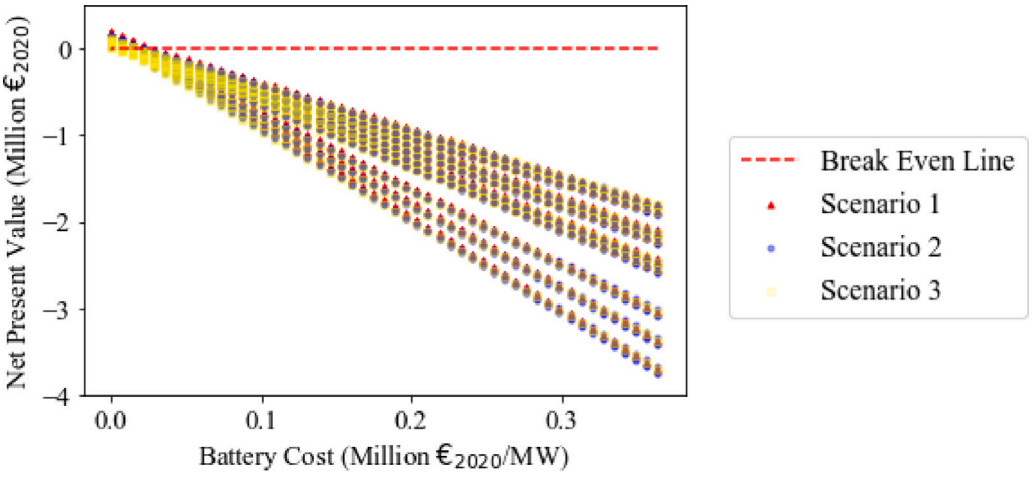


Fig. 8. Results from BPM for Unnån and Hansjö.

from the CHBS when compared to other flow scenarios. As the flow increases from medium to high, the discounted cashflow increases by a very small margin. This can be attributed to the restrictions imposed by the upper limit on injecting power back into the grid. Similar to Skattungbyn, the results from changing the flow in Unnån and Hansjö have the same trend as observed in Table 18.

The results from different cases for the battery in Skattungbyn for short term regulation until the year 2040 is shown in Fig. 7. The Fig. 7 consists of different battery capacities and the price of BESS has been varied from 100% to 0% of the current BESS costs.

From Fig. 7, six different lines can be observed, which corresponds to six different BESS size configurations from Table 13. It can be observed in Fig. 7 that the NPV is negative for most of the cases. As the battery costs per MW are varied from its current costs 100% to 0%, the NPV increases as the investment decreases. There is not much of a difference between the NPV among the different scenarios, but scenario 1 has a slightly higher NPV than scenarios 2 and 3. Scenario 3 has the least NPV among the price scenarios. A break-even line is also seen in the graph, which depicts when the investment is either profitable or note. The intersection of this line typically happens when the battery cost is less than 40 000 € / MWh, which is 90% less than the current cost of the battery (which is 0.36 Million € / MWh).

Similarly, the results from BPM for a battery for both Unnån and Hansjö together for short term regulation until the year 2040 is shown in Fig. 8.

The cost of the battery to achieve the breakeven point in this location is around 50 000 € / MWh. The NPV is less for the battery in Unnån and Hansjö when compared to the NPV for the battery in Skattungbyn. From both the graphs, it can be observed that the NPV is negative for the current battery costs and thus concluding that short term regulation in these hydropower plants is not profitable. The battery costs must significantly reduce, almost to 12% of its current costs, to attain the breakeven point.

4.4. Discussion

The electricity price has been a difficult entity to forecast over the years. With most of the research on forecasting electricity prices falling in the short term, there has not been a significant amount of research that can be used to compare the results from this paper. The average day-ahead electricity price for 2040, based on a report from April 2022 by European Network of Transmission System Operators for Electricity (ENTSO-E) is around 60 €/MWh based on the weighted average of marginal prices in EU27 countries [50]. In comparison to this, the average day-ahead electricity price predicted by FEPFM for 2040 from Scenario 1, 2 and 3 are 74 €/MWh, 35 €/MWh and 30 €/MWh respectively. Hence, it can be observed that there is some difference in terms of predicted absolute values, but the scenario design plays a huge role in predicting more relevant future prices. The prices of electricity have a daily, weekly, and seasonal pattern over the years. The price of electricity is significantly influenced by production from

different energy sources, hydro reservoir amount and lagged electricity prices. One of the important factors which can improve the long-term forecasting model is including the imports and exports of electricity to and from Sweden. As Sweden being part of a connected electricity market along with other countries, the price of electricity is also influenced by the electricity demand and production from the other countries participating in the same electricity market. But due to the uncertainty in imports/exports and the difficulty to predict it in an hourly resolution from 2020 to 2040, it was not considered in the forecasting model. In addition to this, the FEPFM has some limitations which might help in making the model as close to real-life scenario. The first limitation is that the model considers no improvements in terms of increasing electrical consumption efficiency, heating insulation of building materials' occurrence of natural disasters which might influence the renewable energy production which in turn influences the electricity prices. Furthermore, as the model is based on machine learning, the model predicts the prices based on existing data. As a result of this, FEPFM would not be able to consider the possibility of having large scale storage and smart consumers. Considering the current trend of technological advancement, it is essential to incorporate these technological advances to predict the prices accurately, but has been skipped in this study due to data unavailability and model simplicity.

The most influencing factors on the electricity price from FEPFM, are the load and the wind energy production. The load influences the electricity price directly and the wind energy production indirectly influences the electricity price, which can be viewed from Fig. 4. The Swedish Wind Energy Association predicts that the electricity demand would be around 200 TWh which is roughly 60% increase from the current annual consumption of around 125 TWh [45], whereas the results from North European Power Perspectives, modeled three scenarios and predicted the annual load to be varying between 120 and 200 TWh for the year 2040 [51]. As in this paper, the loads were assumed to reach 200 TWh as stated by [45], the electrical load in Sweden is supposed to be increasing tremendously over the coming years, with the increasing population leading to the increased demand for electricity. ANN yielded the best result in FEPFM with an MAPE of 13.05% which is on par with the results from machine learning as stated in the literature review. The role of nuclear power in the Swedish electricity mix is noteworthy. In Scenarios 1 and 2, the influence of removing nuclear energy can be observed. Nuclear energy acts as a base load in the Swedish electric system and helps bring stability. The complete phase-out of nuclear would cause the price to escalate and makes it more volatile. The prices are supposed to increase in the future with the increase in demand and phasing out of nuclear, and its effect on the electricity price surpasses the effect of RE production in the electricity mix.

The forecasted electricity prices are one of the most critical inputs for techno-economic optimization. In addition to FEPFM, the BPM does pose one limitation: the calendar aging due to SOC of the BESS has been skipped to reduce computational complexity. As discussed in Section 3.1.1, multiple cases were developed based on varying parameters. The parameters which were varied were battery cost, battery capacity, battery connection point, changing the grid agreement and electricity prices from different scenarios of FEPFM. From analyzing the data points in the NPV graphs plotted in Figs. 7 and 8, it was observed that the change in battery cost has the highest effect on the NPV followed by battery size, changing the grid agreement, electricity price scenarios and flow scenarios. It was observed from the results that the increase in BESS capacity had an inverse impact on NPV as the cost of the BESS increases. It was identified that grid agreement limits the battery size as it determines the amount of power injected into the grid, which further limits the power that can be sold in the electricity market to generate revenue. The results from this paper are in line with the results from [9,41] which suggested that the addition of batteries to hydropower plant is still not feasible, and the battery costs should reduce significantly to make it a valuable investment. Technical limitation from the electrical system and the cost for increasing the grid, agreement with the local grid operator limits the battery size and operation as well. The results from the economic analysis show that the current costs of new batteries are significantly great, and implementing such batteries in the hydropower plants in lower Oreälven is not profitable for short-term regulation. The main limitation of implementing the batteries are electrical restrictions posed by the grid agreements with the grid operator, high capital costs of the batteries and the size of the power plants. The grid agreements can be altered, but it incurs a significant cost, which is subsequently removed from the revenues from the CHBS. The power plants in focus are small and have a relatively small capacity. This restricts the BESS operation. Also, the power plants in focus are operated like RoR hydropower plants as if they had no reservoir storage, due to the water rights permit limitations for Skattungen. If the electricity agreements can be altered without incurring costs, then the CHBS would yield high returns. In order to maximize the returns from the CHBS, other possibilities of employing the CHBS in other markets must be explored. The model works only when there is sufficient difference in electricity price within the horizon of optimization. There can be opportunities when the battery could be used in regulation markets during times when difference elspot prices are not significant for arbitrage. But in order to explore such options, a detailed analysis on the regulation market must be first explored along with future prediction of regulation market prices. This demands a separate detailed study on the regulation market in order to explore this opportunity.

In addition to this, the developed model has possibility to be scaled up and replicated for other regions given the availability of the data. Once the forecasted prices and the techno-economic data regarding the hydropower plants are known, the proposed model (FEPFM and CHBS) can be used to analyze the combined operation of hydropower plants with storage in other regions.

5. Conclusion

The increasing attention to further environmental flow restrictions have reduced the potential profits of hydropower plants by limiting the flexibility in operation. One potential solution around this would be the possibility to incorporate a storage system with the hydropower plant and short-term regulate the combined system. To study this, scheduling using MILP optimization of hydropower plants has been proposed, considering modeling of the BESS. Machine learning based ANN model to predict the hourly wholesale electricity prices have been proposed as the first stage of the method. The mentioned ANN model has an MAPE of 13% considering the data from 2013–2019, which is used to predict the prices from 2020 to 2040. To study the performance of the proposed models, a case study on hydropower plants in lower Oreälven in Sweden were used. The feasibility of short-term battery regulation in the three hydropower plants in was analyzed both in technical and economic terms, considering environmental constraints, in this paper. Various cases were studied and analyzed by varying the cost of the battery, size of the battery, connection point of the battery, grid agreement extension and future electricity market scenarios. The key findings from this study are:

- The paper forecasted the hourly wholesale electricity prices from 2020 until 2040 based on three different scenarios of future energy mix considering phasing out of nuclear power in Sweden.
- The predicted electricity prices were inline with the expected annual average electricity from ENTSO-E [50].
- The NPV of implementing the BESS in the hydropower plants in Oreälven is negative (from −1 to −4 Million € based on different scenarios) for the current battery costs obtained from NREL [49].
- The effect of CAPEX of BESS is one of the limiting reasons for the negative NPV, while additional reasons being grid limitations and the small size of the hydropower plants.

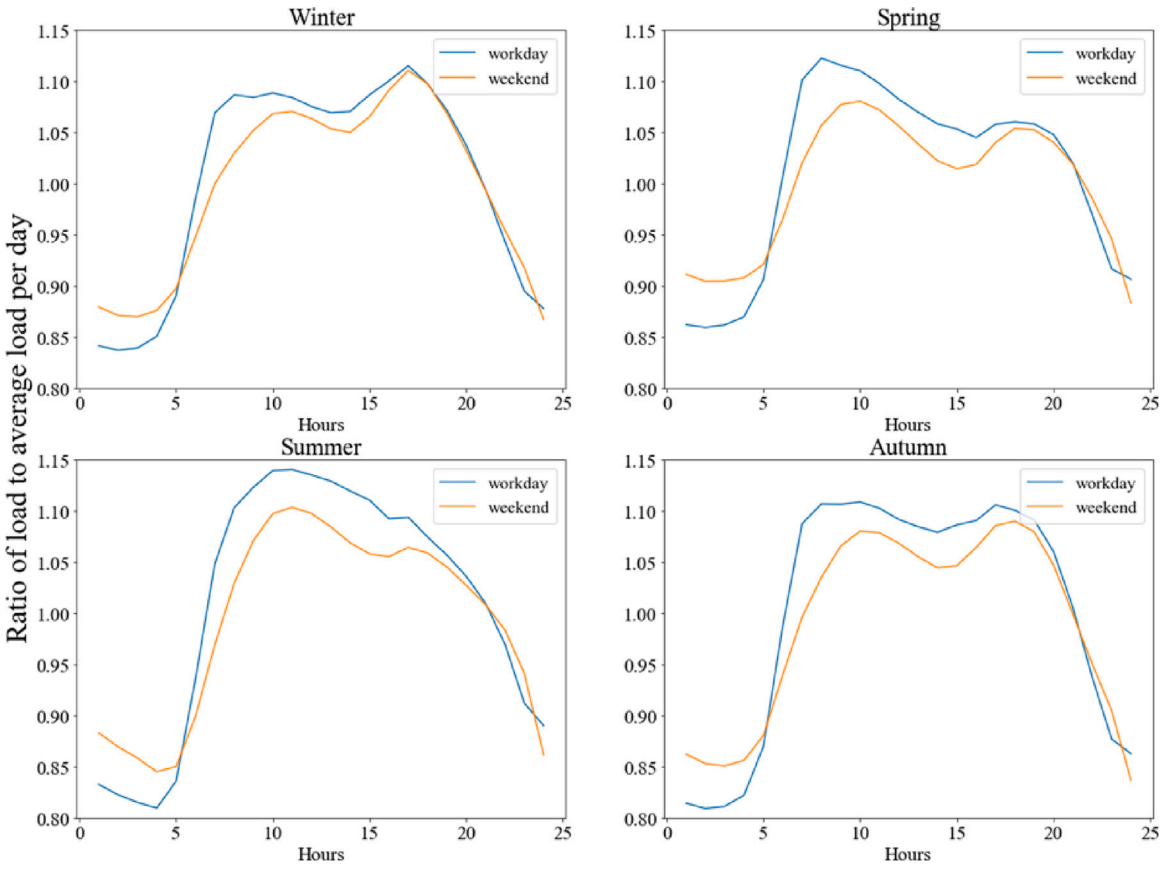


Fig. A.9. Ratio of the load to the average load per day in different seasons.

Through this study, it was possible to analyze the potential returns of incorporating a BESS in the hydropower plants for short-term regulation. However, it is important to consider the limitations posed by this study. The uncertainty in prices and future improvements in efficiency is not considered. Though the short-term regulation in the day-ahead wholesale market is not profitable, the proposed combined operation of CHBS can be used in other services within the electricity markets (for e.g. regulation and intraday markets) to maximize the potential returns. The study of such markets and prediction of prices for these alternatives is proposed for future studies. In addition to this, the effect of alternative storage technologies should be studied and the influence of such technologies on the electricity prices can be considered for future research.

CRediT authorship contribution statement

Araavind Sridhar: Conceptualization, Methodology, Software, Validation, Data curation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Ashish Guhan Baskar:** Conceptualization, Methodology, Software, Validation, Data curation, Resources, Writing – original draft, Writing – review & editing, Visualization. **Jagruti Thakur:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration, Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Future input prediction for FEPFM

A.1. Electrical energy consumption (load)

Sweden has been a net exporter of electric energy. The imports and exports of Sweden depend on the availability or lack of electricity in all the countries which are interconnected to Sweden for electricity. It is based on the policies for energy development in these interconnected countries, and it is hard to determine an exact value of how much imports/exports Sweden would have in the future. With no proper trend being observed from historical data and in order to not add more uncertainty in the model, an average of export from 2013 to 2019 which is 8.98% is fixed as an assumption and based on it, the load is predicted in the future. The total production for 2040 is calculated as 208 TWh based on reports from Swedish Wind Energy Association [45]. Hence, the load for the year 2040 is 198 TWh with a net 8.98% export. considering a linear increase from the years 2019 to 2040 the total load every year is identified. The data of the load is calculated on an hourly basis. To estimate load for every hour in the future, a baseline load profile is estimated based on different seasons and average load throughout 2013 to 2019 and is shown in Fig. A.9.

This baseline annual load is then scaled up accordingly based on the total load requirement in different years based on different scenarios.

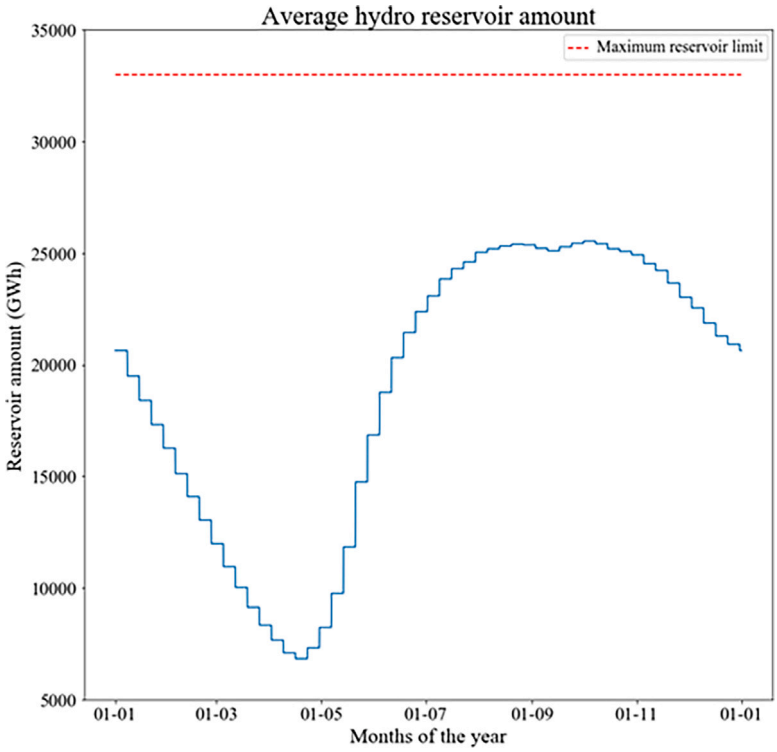


Fig. A.10. Average hydro reservoir amount in the future.

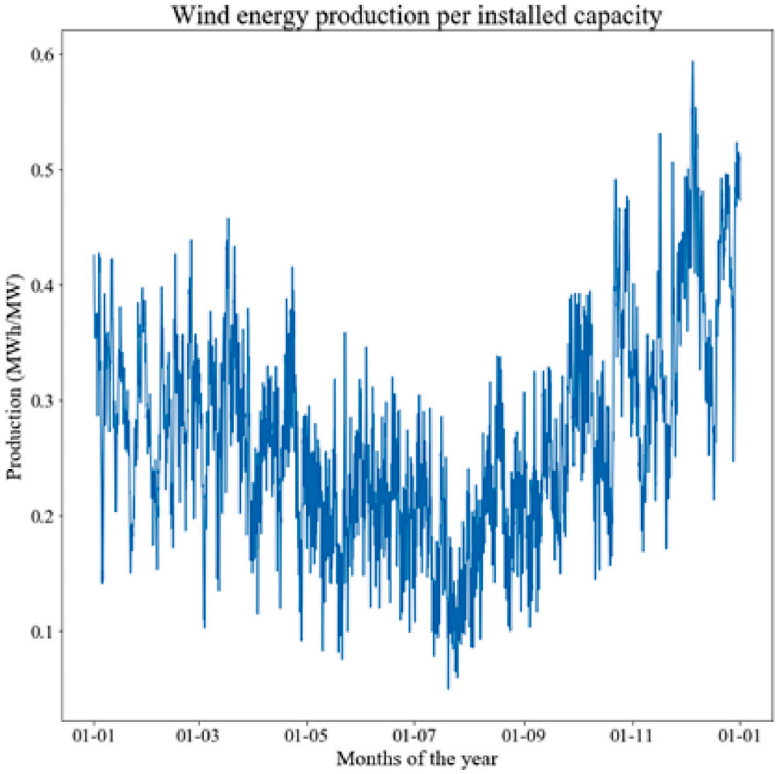


Fig. A.11. Wind energy production for future—-Base profile.

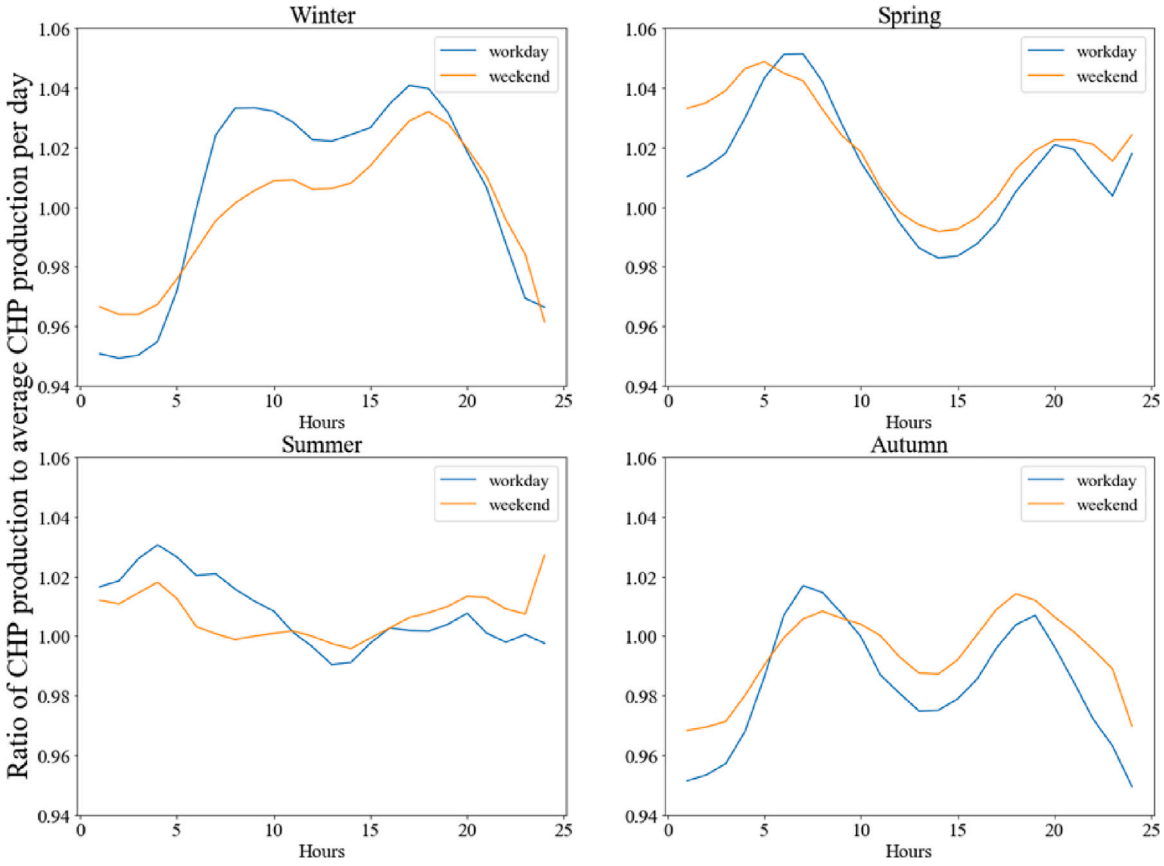


Fig. A.12. Ratio of CHP production to average CHP production per day in different seasons.

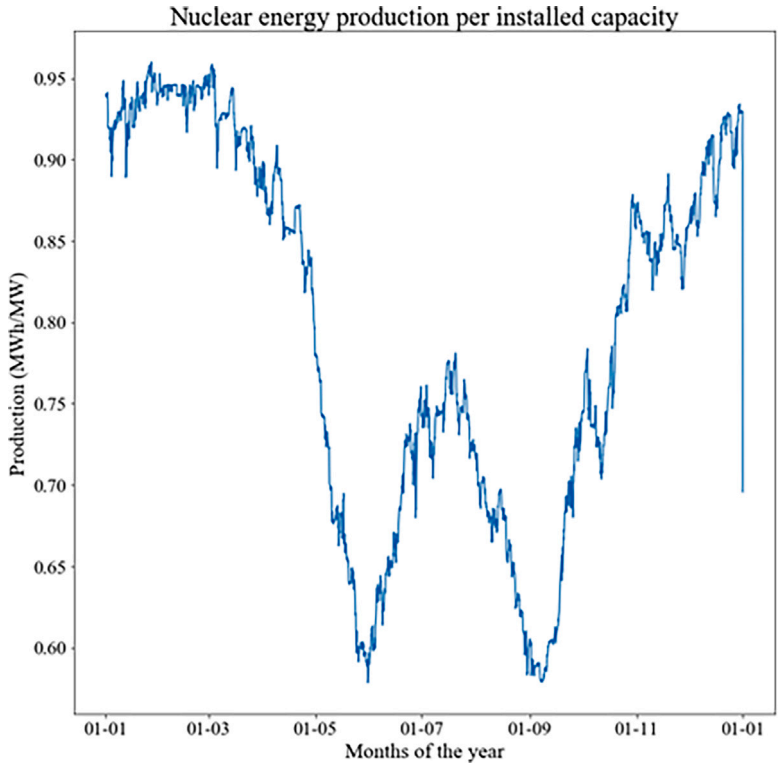


Fig. A.13. Nuclear energy production per installed capacity—nuclear base profile.

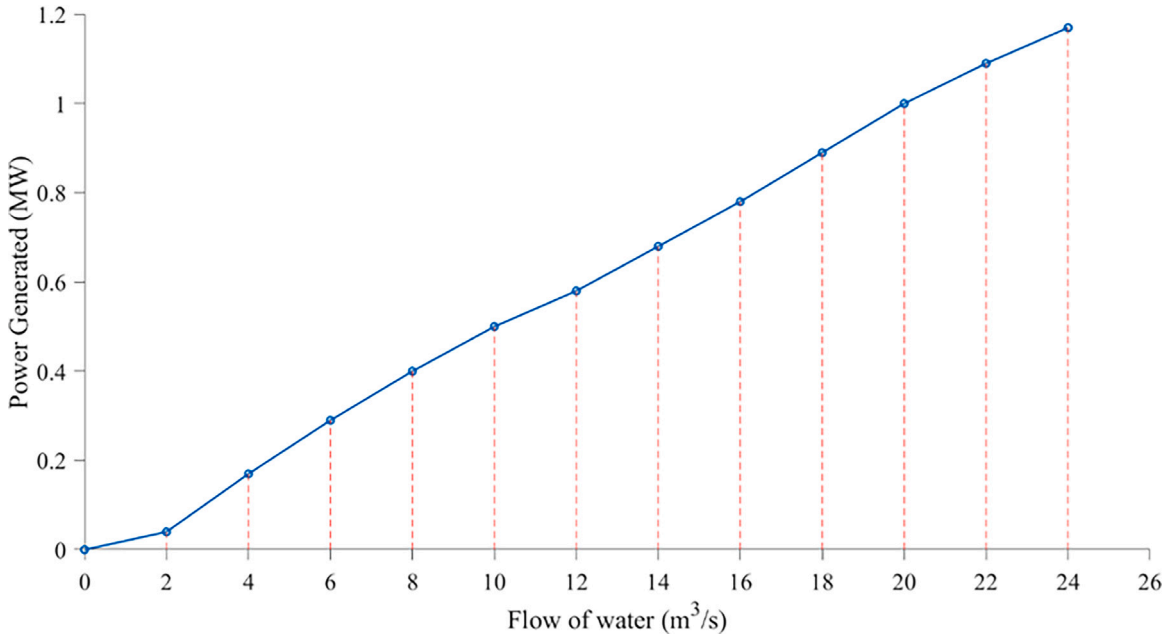


Fig. B.14. Piece-wise linear function for the Kaplan turbine in Skattungbyn power plant.

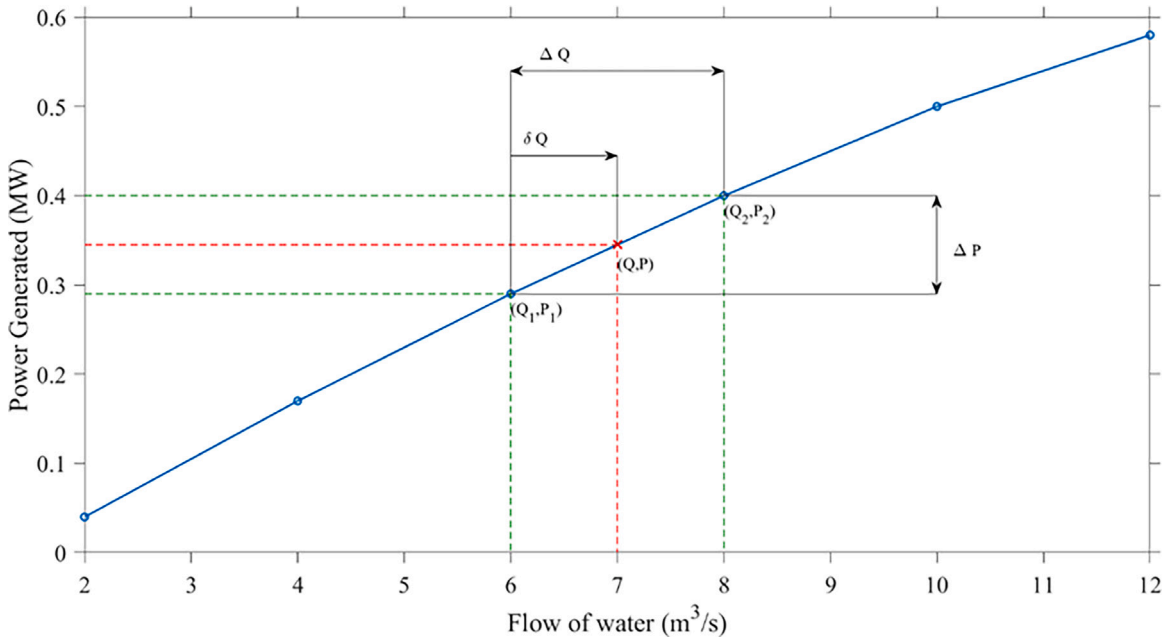


Fig. B.15. Point selection in the piece-wise linear function.

A.2. Hydro reservoir amount

It represents how much hydropower can be produced based on the capacity of the hydropower plants and water availability. It is measured every week and assumed to be the same throughout the week. This factor is assumed to be constant throughout the years because it is influenced by rain, snow, temperature, and production from other sources of energy. An average value of hydro reservoir amount from 2013 to 2019 was considered as an input for FEPFM as shown in Fig. A.10. This data is collected on a weekly basis.

A.3. Wind energy production

The amount of wind power plants installed every year is identified and the ratio of wind energy produced every hour to the installed capacity would give a baseline profile that can be used to scale it up in the future based on the predicted installed capacity in Sweden [45]. The power factor of wind power is assumed to be constant based on the historical data as shown in Fig. A.11. The data of wind energy production was calculated on hourly basis.

A.4. Biomass CHP production

CHP powerplants in Sweden are operated based on the heat demands, and hence the generation is high during the winter months and low during the summer months. CHP production pattern is observed to be close to the electric load. Hence, an identical approach that was carried out to predict the future load is followed to predict future CHP production, as shown in Fig. A.12. The data of biomass CHP production was calculated on hourly basis.

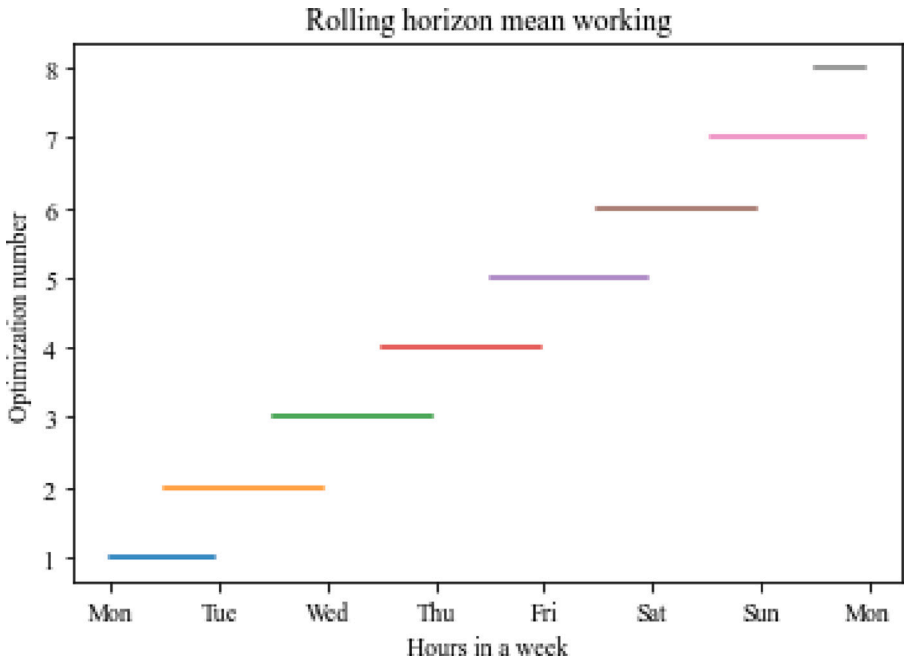


Fig. C.16. Working of rolling horizon in BPM.

A.5. Nuclear energy production

The nuclear power plants are operated around their full capacity from the end of autumn to the beginning of spring. There are reserved planned hours for the maintenance of nuclear power. A similar approach to that of predicting wind energy is used to assess nuclear energy production, as observed in Fig. A.13. The data of nuclear energy production was calculated on hourly basis.

Appendix B. Piecewise-linearization

The electrical power generation in the subsystem is dependent on the water flow in the turbine. The relation between different flow points and electrical power is defined by a flow vs power graph for the turbine and is turbine specific. The data points are determined by testing the turbine and acquiring data from measurement devices. An equation is obtained by linear curve-fitting on the data points, which is used to compute the power production for any flow in the given range of operation. The best-fit curve equation from the graph is linear, which might not pass through all the data points. To compute the power output for different flow points more accurately, a piecewise linear function is used. In this, the data gap between two known data points is considered as a linear curve as shown in Fig. B.14, and for any point which lies between the known data points, the corresponding power for the flow value can be computed with the linear equation for that data interval.

To understand the functioning of a piecewise linear function, random flow of water is considered. In the example shown in Fig. B.15, the flow data is considered to be $Q = 7\ m^3/s$ and the P is to be determined. This point lies between the known data points Q_1, P_1 (6 m^3/s) and Q_2, P_2 (8 m^3/s). A linear curve with slope "m" is created between the points Q_1, P_1 and Q_2, P_2 by using the two-point theorem. The slope of the linear curve is $m = (P_2 - P_1)/Q_2 - Q_1$ P value for the corresponding Q is found by $P = p_1 + (Q_2 - Q_1) * m$ method is used to find any operating point of the turbine.

Appendix C. Rolling horizon optimization

In order to understand the working of BPM, the explanation of rolling horizon is important. To explain its working, Fig. C.16 can be observed. In this figure, the x axis has the hours of the week, stating from Monday until Sunday. The y axis shows the optimization number, which is computed by the BPM.

From Fig. C.16, the BPM initially optimizes the CHBS to operate based on the prices for the whole of Monday, which is represented by the blue line in the figure. But when the day-ahead electricity prices for Tuesday, are published just before hour 13 on Monday, the BPM recalculates the operation of the CHBS from hour 13 on Monday until the end of Tuesday, which is represented by the orange line in the figure. Similar to this, on Tuesday just before hour 13, the prices for Wednesday are known. Therefore, the BPM optimizes the CHBS again from 13th hour of Tuesday until the end of Wednesday. This process keeps on repeating until the end of the operation time horizon.

Appendix D. Dynamics of the CHBS

Fig. D.17 shows the dynamics and working of the combined CHBS. There are three different flow in terms of water flow, power flow from turbine and power flow from the battery. The water flows from river upstream to river downstream as seen from the blue arrow in the figure, wherein it goes through the hydropower plant. The hydropower plant is composed of a spillway and turbine flow. The flow in the turbine produces electricity and the flow through the spillway does not. Once the electricity is produced from the turbine, it is then transmitted to the grid and/or the battery, which can be depicted by the red arrow in the figure. The battery can charge from the turbine and discharge to the grid, as observed from the green arrow in Fig. D.17.

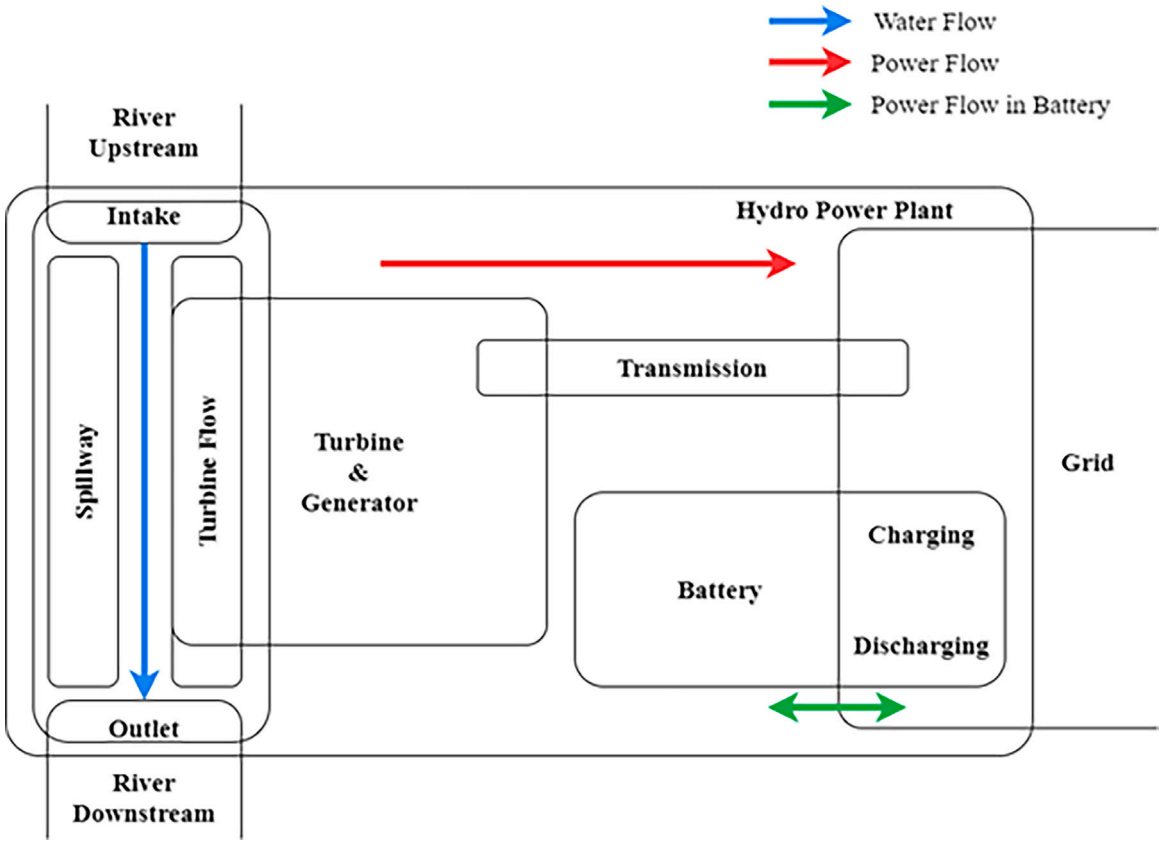


Fig. D.17. Working of CHBS.

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